

Do **not** write solutions on this page.

12. [Maximum mark: 20]

Consider the equation $z^4 = 16i$, where $z \in \mathbb{C}$.

The equation has four roots z_1, z_2, z_3, z_4 , where $z_i = r(\cos \theta_i + i \sin \theta_i)$, $r > 0$ and $0 \leq \theta_1 < \theta_2 < \theta_3 < \theta_4 < 2\pi$.

- (a) Find z_1, z_2, z_3 and z_4 . [6]

The roots z_1, z_2, z_3 and z_4 form a geometric sequence.

- (b) Find the common ratio of the sequence, expressing your answer in Cartesian form. [3]

The roots z_1, z_2, z_3 and z_4 are represented by the points A, B, C and D respectively on an Argand diagram.

- (c) Plot the points A, B, C and D on an Argand diagram. [3]

The equation $v^4 = a + bi$, where $v \in \mathbb{C}$ and $a, b \in \mathbb{R}$ has roots z_1^*, z_2^*, z_3^* and z_4^* .

- (d) Determine the value of a and the value of b . [3]

The midpoint of [AB] is A' , the midpoint of [BC] is B' , the midpoint of [CD] is C' and the midpoint of [DA] is D' .

Consider the equation $w^p = 2^q$, where $w \in \mathbb{C}$ and $p, q \in \mathbb{Z}^+$.

Four of the roots of $w^p = 2^q$ are represented by the points A', B', C' and D' .

- (e) Find the least possible value of p and the corresponding value of q . [5]

